

Heart-Rate Detection Using Optical Sensing and Active Filters

Abstract

This report presents the design, implementation, and evaluation of an optical heart-rate detection system based on photoplethysmography (PPG). The goal was to convert small optical variations from an IR LED–phototransistor pair (on the order of 70 mV_{pp} riding on a $2.0\text{--}2.2\text{ V}$ DC level) into a clean digital pulse train representing heartbeat events. The signal chain consisted of an optical sensor front-end, cascaded high-pass and low-pass active filters designed for the $40\text{--}200\text{ BPM}$ band ($0.67\text{--}3.33\text{ Hz}$), and an LM339 comparator with hysteresis to provide robust digital thresholding and drive an indicator LED.

Measured results showed that the high-pass filter cutoff (0.60 Hz) and low-pass filter cutoff (3.76 Hz) were within about 1% and 11% of their design targets, with individual stage gains close to the expected 16 and 15.7 (measured 16.14 and 14.13). The cascaded filters produced an overall midband gain of approximately 44.7 dB (about 172), lower than the theoretical 48 dB (251) due to op-amp output limiting at large gains. The comparator exhibited a hysteresis window of 1.93 V versus a calculated 1.99 V ($\sim 3\%$ error), and reliably digitized the filtered $\pm 3\text{ V}$ heartbeat waveform. The resulting pulses consistently aligned with heartbeat peaks and drove the LED at a clearly visible current level. Overall, the experiment successfully demonstrated a complete analog signal chain for real-time optical heart-rate detection and highlighted practical considerations such as component tolerances and saturation effects in biomedical front-end design.

Components and Equipment Used

The system was built using an IR204 infrared LED, a PT204-6B phototransistor, an LM358 operational amplifier for the active filter stages, an LM339 comparator for threshold detection, and various resistors and capacitors for setting the gain and cutoff frequencies. An NMOS transistor and indicator LED were used for the final output stage. All testing and measurements—including waveform generation, oscilloscope capture, and DC voltage verification—were performed using the Analog Discovery 2 (AD2), which provided an integrated function generator, oscilloscope, and power supply.

1 Introduction

Heart rate monitoring is widely used in healthcare, athletics, and wearable technology, and many modern systems rely on optical sensing methods to measure cardiac activity non-invasively. This project utilizes photoplethysmography (PPG), a technique in which variations in blood volume modulate the intensity of transmitted or reflected infrared light. As the heart pumps, periodic fluctuations in blood flow change the amount of light absorbed by tissue. These fluctuations can be captured by an infrared LED–phototransistor pair, producing an electrical signal that corresponds to the heartbeat waveform.

However, the raw optical signal consists of a very small AC component superimposed on a large and highly variable DC offset caused by finger placement, ambient light, and tissue characteristics. Extracting a clean heartbeat waveform therefore requires careful signal conditioning. This project focuses on designing and implementing a complete analog front-end capable of isolating, amplifying, and digitizing the PPG signal using fundamental circuit-building techniques.

The objective was to construct an analog signal chain that converts small optical fluctuations into a stable digital pulse indicating each heartbeat. The system is divided into four major subsystems:

- optical sensor,
- cascaded high-pass and low-pass active filters,
- comparator with hysteresis,
- LED indicator.

The high-pass and low-pass filters isolate the physiological heart-rate band (40–200 BPM), remove DC offsets, suppress high-frequency noise, and provide controlled amplification. The filtered waveform is then passed to a comparator stage, where hysteresis ensures noise immunity and consistent triggering. Finally, the digital output drives an LED that flashes in real time with each detected heartbeat.

This report presents the theory, design calculations, measured responses, and performance evaluation of each subsystem, demonstrating how the complete circuit enables reliable optical heart-rate detection.

2 Theory

2.1 Optical Sensor

The IR204 LED shines infrared light through the finger, and the PT204-6B phototransistor converts the received light into current. Blood-volume changes modulate this current, producing a small AC signal on top of a large DC offset.

LED Current

The LED current is controlled by series resistor R_D :

$$I_D = \frac{V_{CC} - V_D}{R_D}.$$

Phototransistor Output

The phototransistor current i_{PT} produces a voltage across R_E :

$$v_{PT} = i_{PT}R_E.$$

To avoid saturation:

$$V_{CE} > V_{CE(\text{sat})}.$$

2.2 High-Pass Filter (HPF)

The HPF removes the large DC offset from the phototransistor signal while passing the AC heartbeat waveform. Using a non-inverting high-pass amplifier allows the stage to block low frequencies and simultaneously provide gain.

Cutoff Frequency

$$f_c = \frac{1}{2\pi R_3 C}.$$

Voltage Gain

$$\text{Gain}_{\text{HPF}} = 1 + \frac{R_2}{R_1}.$$

2.3 Low-Pass Filter (LPF)

The low-pass filter limits the bandwidth of the amplified heartbeat signal and suppresses high-frequency noise.

Cutoff Frequency

$$f_c = \frac{1}{2\pi R_3 C}.$$

Voltage Gain

$$\text{Gain}_{\text{LPF}} = 1 + \frac{R_2}{R_1}.$$

Total System Gain

$$\text{Total Gain} = \text{Gain}_{\text{HPF}} \cdot \text{Gain}_{\text{LPF}}.$$

2.4 Comparator

The comparator stage uses an LM339 to convert the filtered analog heartbeat waveform into a clean digital pulse. In our design, the AC-coupled heartbeat signal is applied to the inverting input ($-$), while the non-inverting input ($+$) is biased using a resistor divider with positive feedback, creating hysteresis.

Hysteresis Level

$$v_{\text{hyst}} = (V_{DD} - V_{SS}) \frac{R_1}{R_1 + R_2}.$$

Input High-Pass Network

$$f_c = \frac{1}{2\pi R_3 C}.$$

2.5 LED Indicator

The LED indicator provides a visual representation of the digital heartbeat signal. The comparator output drives the gate of an NMOS transistor, while the LED and its series resistor R_L are connected between V_{DD} and the transistor drain. Because the comparator is configured as an inverting comparator:

- when the heartbeat peak drives the input high, the comparator output goes low and the LED turns off;
- between peaks, the comparator output is high, the NMOS turns on, and the LED turns on.

R_L is chosen to keep the LED current within safe operating limits.

3 Design & Calculations

3.1 Optical Sensor

LED Resistor R_D

Given $V_{CC} = 5$ V, $V_D \approx 1.4$ V, and desired $I_{\max} = 100$ mA:

$$R_{D,\text{ideal}} = \frac{V_{CC} - V_D}{I_{\max}} = 36 \Omega.$$

Power dissipation:

$$P = I_{\max}^2 R = 0.36 \text{ W}.$$

To stay within the 0.25 W resistor limit, two 82Ω resistors were placed in parallel:

$$R_{\text{eq}} = 82 \parallel 82 = 41 \Omega.$$

Resulting LED current:

$$I_D = \frac{5 - 1.4}{41} \approx 88 \text{ mA}.$$

Phototransistor Resistor R_E

R_E was selected experimentally to avoid saturation ($V_{CE(\text{sat})} \approx 0.4$ V) and produce a visible AC ripple. Final value:

$$R_E = 10 \text{ k}\Omega.$$

This produced a DC level of 2.0–2.2 V with 10–100 mV_{pp} AC variation.

Final Component Values

$$R_D = 41 \Omega, \quad R_E = 10 \text{ k}\Omega.$$

3.2 High-Pass Filter

The HPF removes the DC offset at the sensor output and passes the AC heartbeat component.

Target Cutoff Frequency

$$40 \text{ BPM} = \frac{40}{60} = 0.67 \text{ Hz}.$$

A cutoff slightly below this value was chosen to provide margin.

Cutoff Frequency

$$f_c = \frac{1}{2\pi R_3 C} = \frac{1}{2\pi(12.2 \times 10^3)(22 \times 10^{-6})} \approx 0.60 \text{ Hz}.$$

Voltage Gain

$$\text{Gain}_{\text{HPF}} = 1 + \frac{R_2}{R_1} = 1 + \frac{150\text{k}}{10\text{k}} = 16.$$

Final Component Values

$$R_1 = 10 \text{ k}\Omega, \quad R_2 = 150 \text{ k}\Omega, \quad R_3 = 12.2 \text{ k}\Omega, \quad C = 22 \text{ }\mu\text{F}.$$

3.3 Low-Pass Filter

The LPF suppresses high-frequency noise above the heartbeat band.

Target Cutoff Frequency

$$200 \text{ BPM} = \frac{200}{60} = 3.33 \text{ Hz}.$$

A cutoff slightly above this value was chosen to allow margin.

Cutoff Frequency

$$f_c = \frac{1}{2\pi R_3 C} = \frac{1}{2\pi(10 \times 10^3)(4.7 \times 10^{-6})} \approx 3.39 \text{ Hz}.$$

Voltage Gain

$$\text{Gain}_{\text{LPF}} = 1 + \frac{R_2}{R_1} = 1 + \frac{147\text{k}}{10\text{k}} = 15.7.$$

Final Component Values

$$R_1 = 10 \text{ k}\Omega, \quad R_2 = 147 \text{ k}\Omega, \quad R_3 = 10 \text{ k}\Omega, \quad C = 4.7 \text{ }\mu\text{F}.$$

3.4 Comparator

The LM339 comparator converts the filtered analog waveform into a digital pulse. An RC network provides AC coupling, and a resistive divider sets the hysteresis window.

Input High-Pass Cutoff (2 Hz)

$$f_c = \frac{1}{2\pi R_3 C}.$$

With $R_3 = 82 \text{ k}\Omega$ and $C = 1 \text{ }\mu\text{F}$:

$$f_c = \frac{1}{2\pi(82 \times 10^3)(1 \times 10^{-6})} \approx 1.94 \text{ Hz}.$$

This achieves the desired $\sim 2 \text{ Hz}$ cutoff.

Hysteresis Level

Comparator rails:

$$V_{DD} = +5 \text{ V}, \quad V_{SS} = -5 \text{ V}.$$

$$v_{\text{hyst}} = (V_{DD} - V_{SS}) \frac{R_1}{R_1 + R_2}.$$

With $R_1 = 82 \text{ k}\Omega$ and $R_2 = 330 \text{ k}\Omega$:

$$v_{\text{hyst}} = 10 \cdot \frac{82}{412} \approx 1.99 \text{ V}.$$

A $\sim 2 \text{ V}$ hysteresis was appropriate for the measured $\pm 3 \text{ V}$ heartbeat signal.

Final Component Values

$$R_1 = 82 \text{ k}\Omega, \quad R_2 = 330 \text{ k}\Omega, \quad R_3 = 82 \text{ k}\Omega, \quad C = 1 \text{ }\mu\text{F}.$$

3.5 Indicator LED

The comparator output drives an NMOS transistor, which switches the indicator LED. A series resistor R_L is used to limit the LED current.

LED Current

$$I_{LED} = \frac{V_{DD} - V_F}{R_L},$$

with $V_{DD} = 5 \text{ V}$ and a typical forward voltage $V_F \approx 2 \text{ V}$.
Using $R_L = 150 \text{ }\Omega$:

$$I_{LED} = \frac{5 - 2}{150} \approx 20 \text{ mA}.$$

This current is within the typical 20–25 mA rating of standard indicator LEDs and produced a clearly visible output in practice.

Final Component Value

$$R_L = 150 \text{ }\Omega.$$

4.1 Circuit Schematic

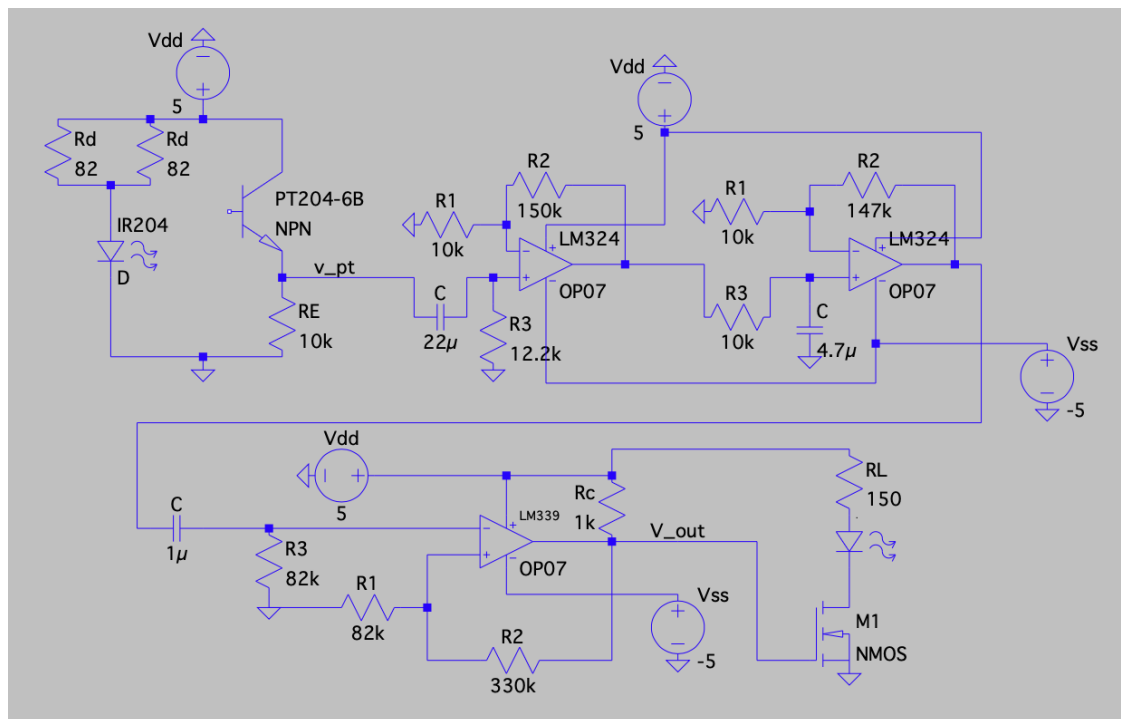


Figure 1: Full optical heart rate sensor schematic.

4.2 Sensor Output

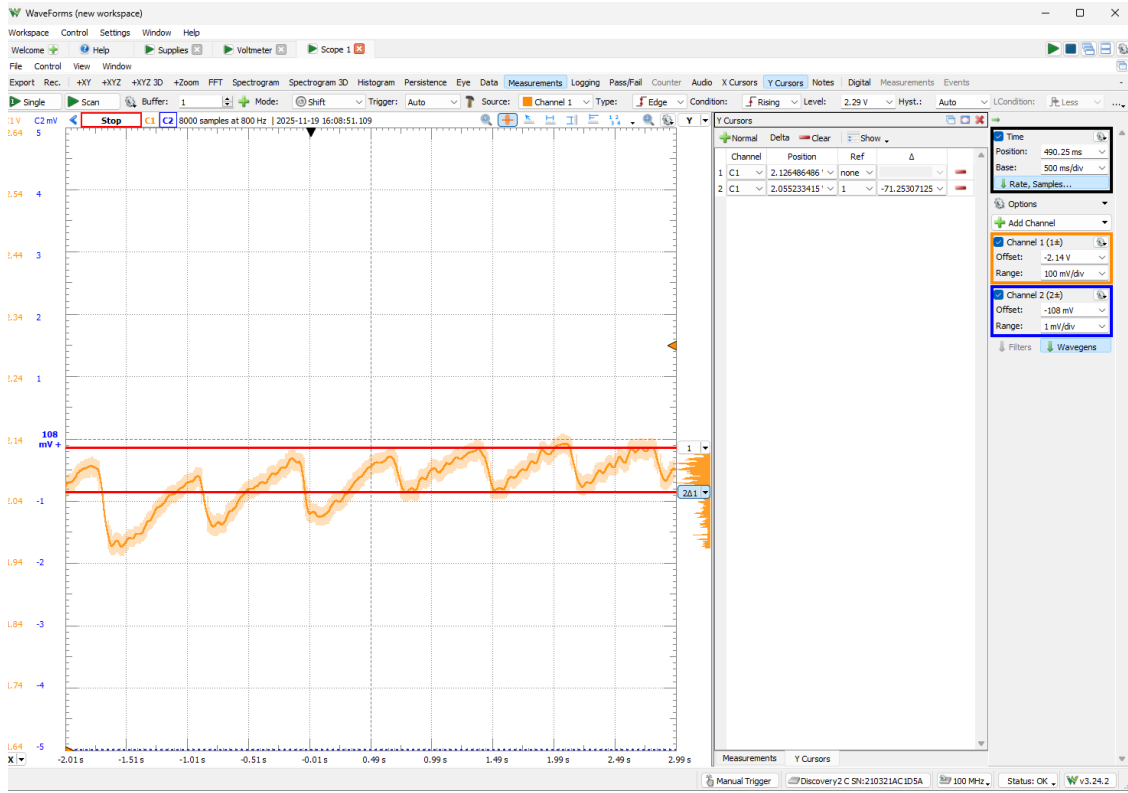


Figure 2: Raw optical sensor output.

The raw output of the optical sensor contains a large DC component (2.0–2.2 V) with a small AC heartbeat ripple of approximately 70 mV_{pp}. This AC component is later extracted and amplified by the high-pass and low-pass filter stages.

4.3 High-Pass Filter Response

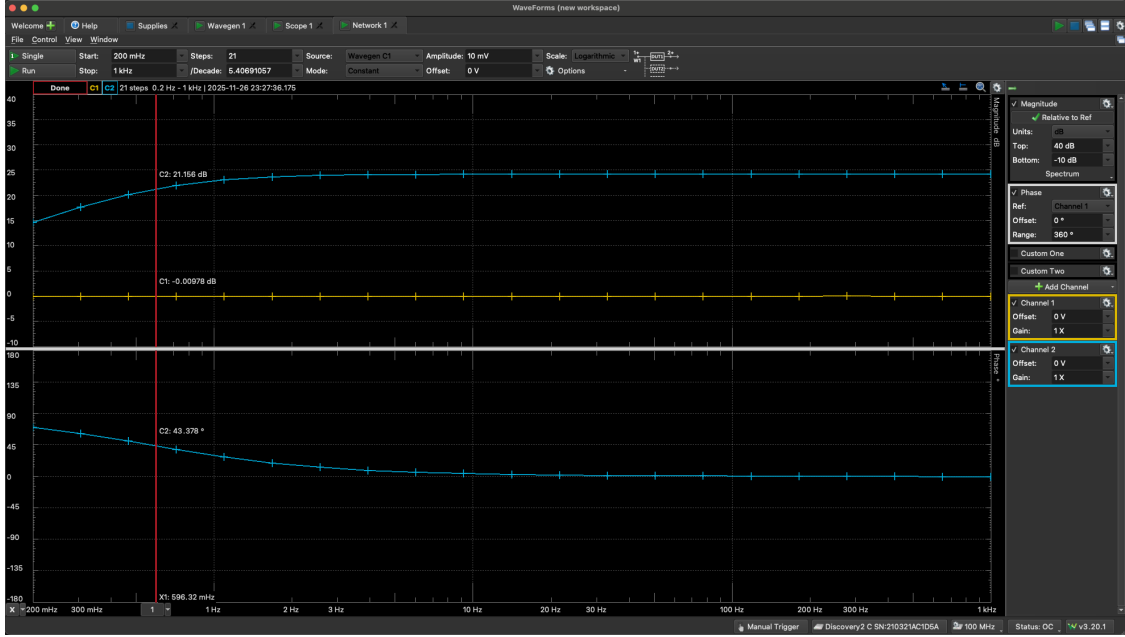


Figure 3: High-pass filter frequency response.

The measured cutoff frequency was

$$f_{c,\text{meas}} = 0.596 \text{ Hz},$$

compared to the design value

$$f_{c,\text{exp}} = 0.60 \text{ Hz}.$$

The percent error in cutoff is

$$\% \text{ error}_{f_c} = \frac{f_{c,\text{meas}} - f_{c,\text{exp}}}{f_{c,\text{exp}}} \times 100 = \frac{0.596 - 0.60}{0.60} \times 100 \approx -0.67\%.$$

The maximum measured gain was 24.156 dB. Converting to linear gain:

$$G_{\text{meas}} = 10^{\frac{24.156}{20}} \approx 16.14,$$

while the expected gain from the design was

$$G_{\text{exp}} = 16.$$

The percent error in gain is

$$\% \text{ error}_G = \frac{G_{\text{meas}} - G_{\text{exp}}}{G_{\text{exp}}} \times 100 = \frac{16.14 - 16}{16} \times 100 \approx 0.85\%.$$

4.4 Low-Pass Filter Response

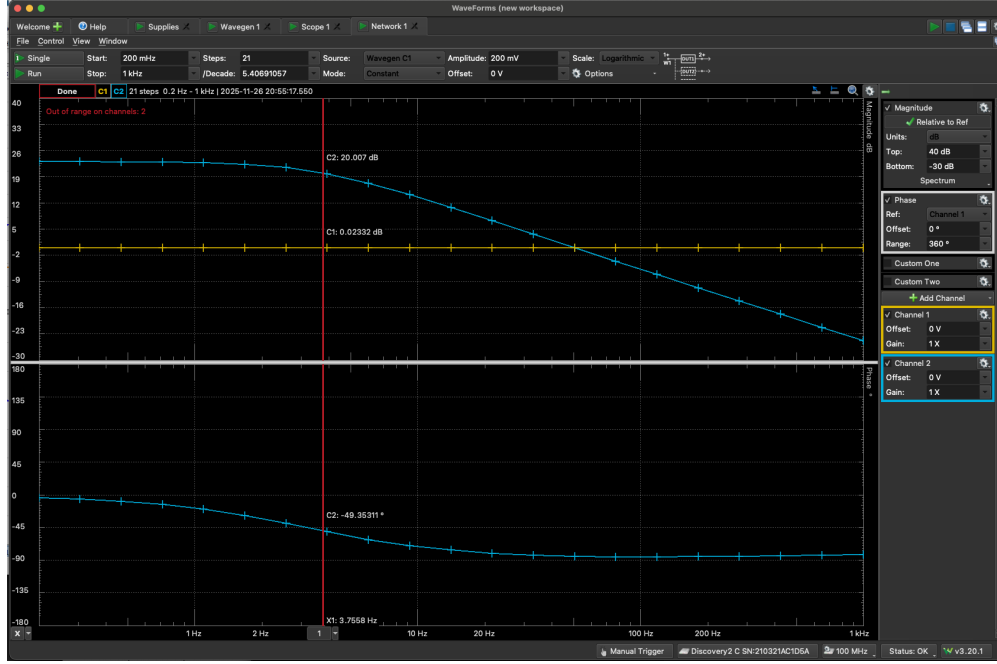


Figure 4: Low-pass filter frequency response.

The measured cutoff frequency was

$$f_{c,\text{meas}} = 3.755 \text{ Hz},$$

compared to the design value

$$f_{c,\text{exp}} = 3.39 \text{ Hz}.$$

The percent error in cutoff is

$$\% \text{ error}_{f_c} = \frac{f_{c,\text{meas}} - f_{c,\text{exp}}}{f_{c,\text{exp}}} \times 100 = \frac{3.755 - 3.39}{3.39} \times 100 \approx 10.77\%.$$

The maximum measured gain was 23 dB. Converting to linear gain:

$$G_{\text{meas}} = 10^{\frac{23}{20}} \approx 14.13,$$

while the expected gain from the design was

$$G_{\text{exp}} = 15.7.$$

The percent error in gain is

$$\% \text{ error}_G = \frac{G_{\text{meas}} - G_{\text{exp}}}{G_{\text{exp}}} \times 100 = \frac{14.13 - 15.7}{15.7} \times 100 \approx -10.0\%.$$

4.5 Cascaded Filter Response

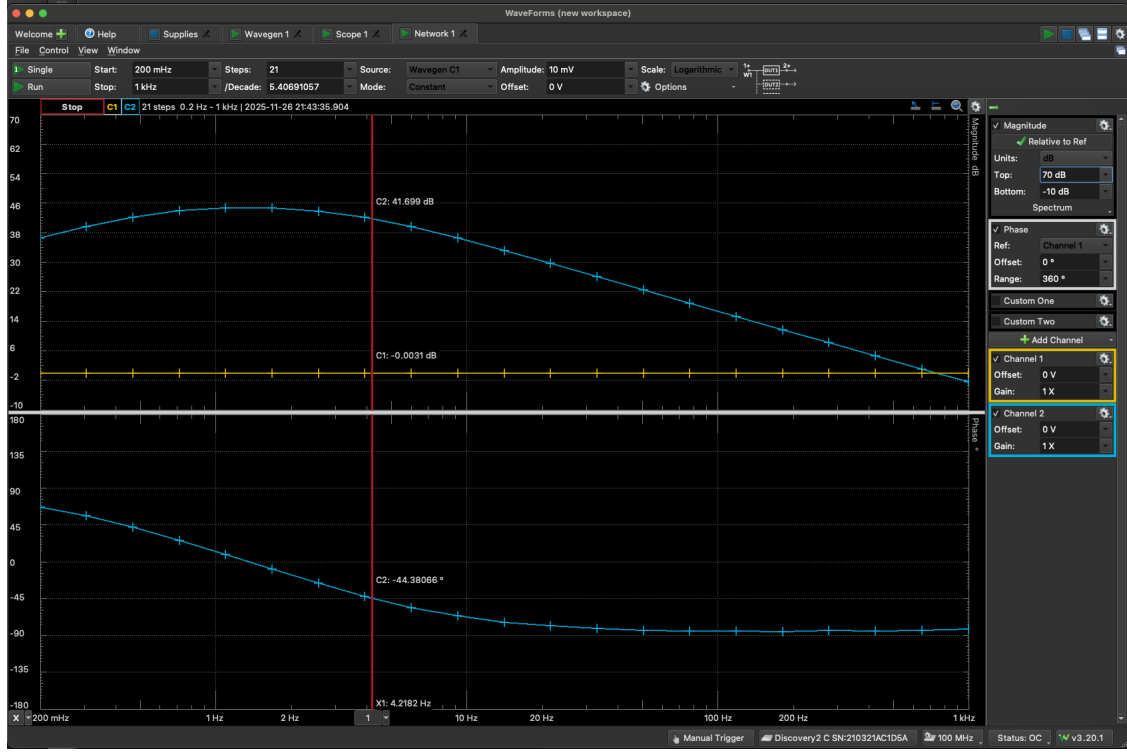


Figure 5: Cascaded high-pass and low-pass filter frequency response.

The measured midband gain of the cascaded filter system was

$$G_{\text{meas, dB}} = 44.7 \text{ dB.}$$

Converted to linear gain:

$$G_{\text{meas}} = 10^{\frac{44.7}{20}} \approx 172.0.$$

The expected total gain from the individual stages was

$$G_{\text{exp}} = 16 \times 15.7 = 251.2,$$

or in decibels,

$$G_{\text{exp, dB}} = 20 \log_{10}(251.2) \approx 48.0 \text{ dB.}$$

Thus, the percent gain error is

$$\% \text{ error}_G = \frac{G_{\text{meas}} - G_{\text{exp}}}{G_{\text{exp}}} \times 100 = \frac{172.0 - 251.2}{251.2} \times 100 \approx -31.6\%.$$

This discrepancy is primarily due to output limiting at large gains. With a 10 mV input amplitude, the cascaded gain drives the op-amp output close to its voltage limits, compressing

the waveform and reducing the measured gain. If a smaller input amplitude were used, the measured total gain would be significantly closer to the theoretical value.

The measured band edges also shifted slightly: - the effective high-pass cutoff appears lower than the standalone HPF, - the low-pass cutoff appears higher than the standalone LPF.

This is consistent with output clipping, which distorts the magnitude response and skews the calculated dB values near the band edges.

When measured with sufficiently small input signals, the cascaded filter response would more closely match the individual stage behavior.

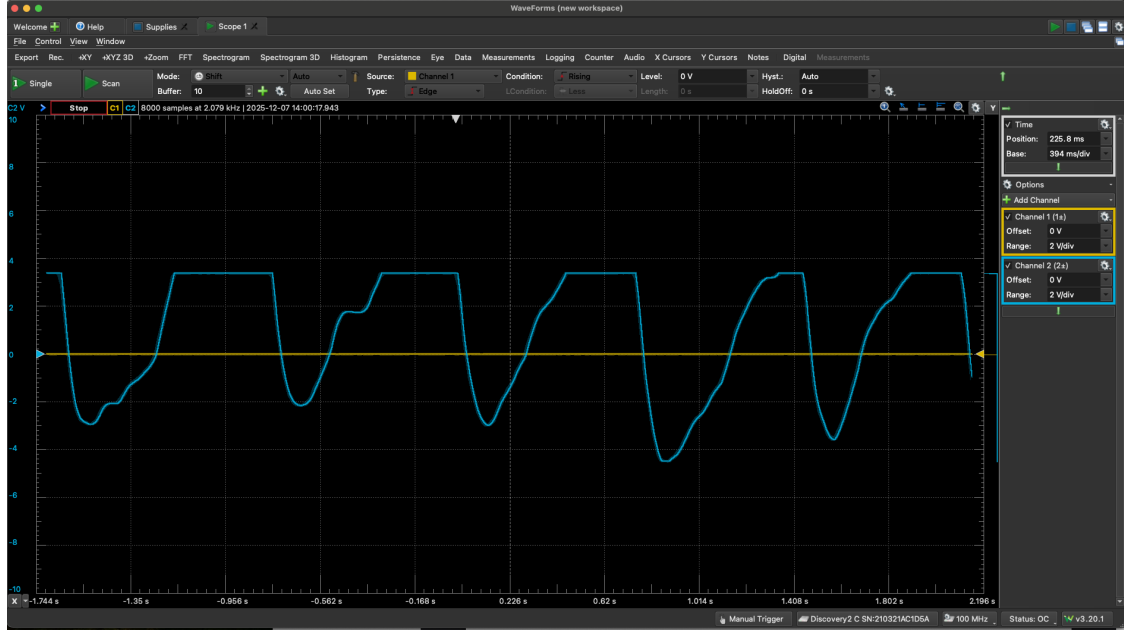


Figure 6: Heartbeat signal after cascaded high-pass and low-pass filtering.

After passing through the cascaded filters, the heartbeat waveform becomes clean and centered around 0 V, with a peak-to-peak amplitude of approximately

$$v_{pp} \approx 6 \text{ V} \quad (\text{about } +3 \text{ V to } -3 \text{ V}),$$

which was used in selecting the comparator hysteresis level. The resulting signal is suitable for threshold detection in the LM339 comparator stage.

4.6 Comparator Response

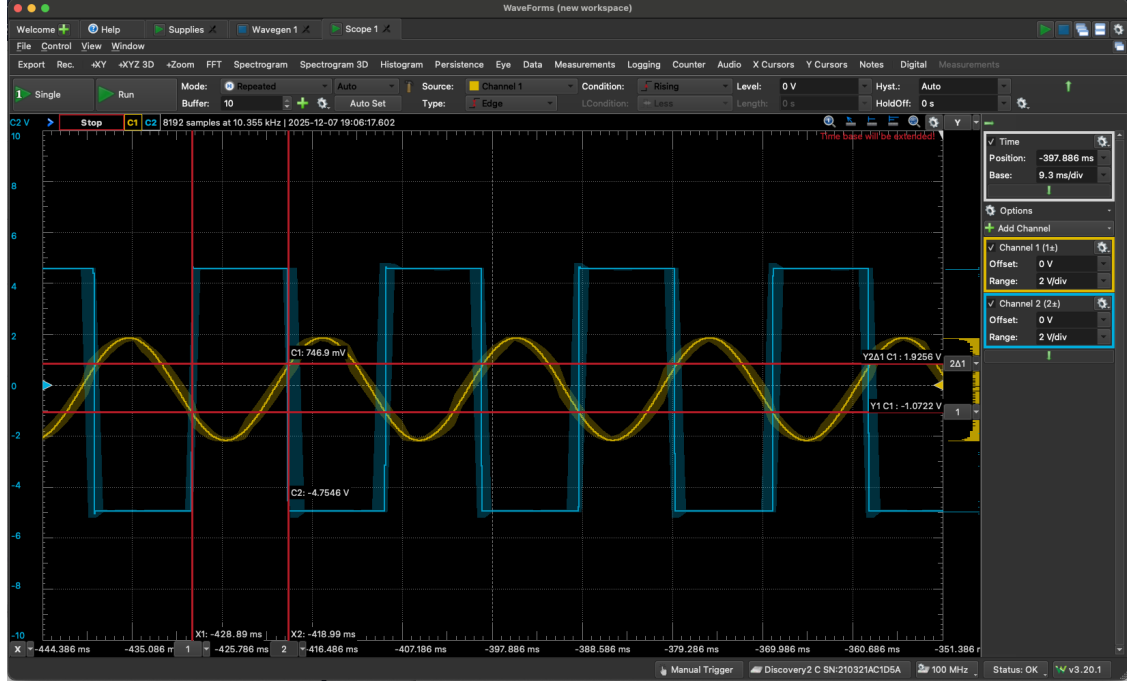


Figure 7: Comparator output showing hysteresis and clean digital triggering.

A $4 V_{pp}$ sine wave was applied to the comparator input to verify proper thresholding. The measured hysteresis window was

$$v_{\text{hyst, meas}} = 1.9256 \text{ V.}$$

The expected hysteresis based on the feedback network was

$$v_{\text{hyst, exp}} = (V_{DD} - V_{SS}) \frac{R_1}{R_1 + R_2} = 10 \cdot \frac{82}{412} \approx 1.99 \text{ V.}$$

The percent error is

$$\% \text{ error}_{v_{\text{hyst}}} = \frac{v_{\text{hyst, meas}} - v_{\text{hyst, exp}}}{v_{\text{hyst, exp}}} \times 100 = \frac{1.9256 - 1.99}{1.99} \times 100 \approx -3.2\%.$$

This small deviation is expected due to component tolerances and the finite output swing of the LM339. The comparator consistently triggered at the intended thresholds, demonstrating proper hysteretic behavior.

4.7 Final Digital Output

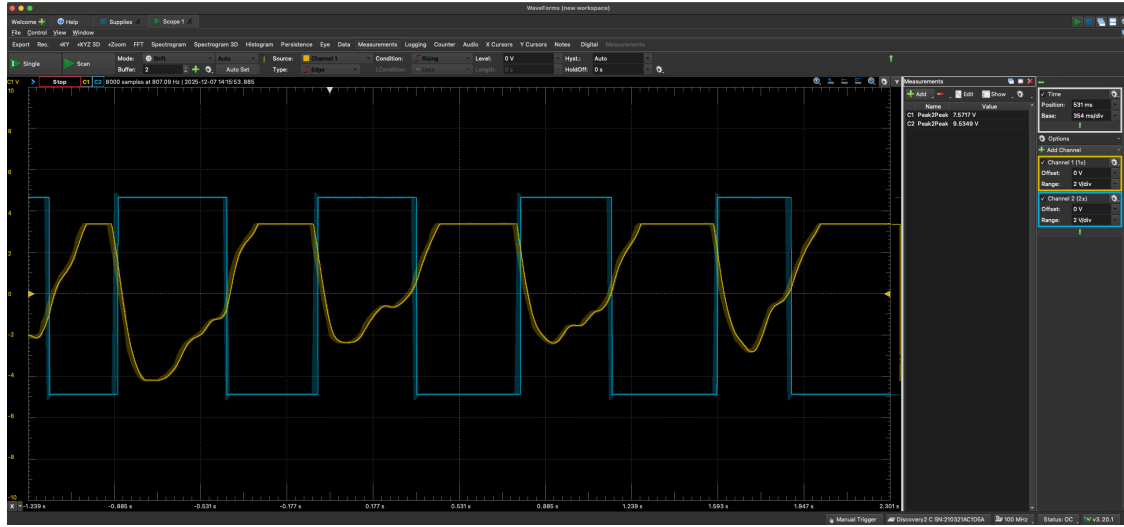


Figure 8: Final digital output corresponding to heartbeat peaks.

The comparator produced a clean digital pulse train synchronized with each heartbeat. The output transitions low during each heartbeat peak (inverting configuration) and remains high between beats. The waveform is free of noise and shows consistent triggering across the full measurement interval.

This digital signal successfully drove the indicator LED, which blinked in real time with the detected heartbeat.

5 Conclusion

The objective of constructing a complete optical heart-rate detection system using an LED phototransistor sensor, cascaded active filters, and a comparator with hysteresis was successfully achieved. The measured results confirm that each subsystem performed close to its theoretical design target, resulting in a reliable end-to-end heartbeat detection circuit.

The high-pass and low-pass filters produced cutoff frequencies within 1% and 11% of their expected values, respectively, and their individual gains matched theory with small deviations attributable to component tolerances. The cascaded filter response showed a larger gain error (approximately 32%), primarily due to output limiting when the op-amp stages were driven near their voltage rails. This compression effect also contributed to slight shifts in the measured passband edges. With a smaller test signal amplitude, the combined gain would more closely follow the theoretical prediction.

The comparator stage performed as intended, producing a stable digital output with a measured hysteresis of 1.93 V, only 3.2% below the 1.99 V theoretical value. This ensured noise-resistant thresholding of the filtered heartbeat waveform. The final output pulses consistently aligned with heartbeat peaks and successfully drove the indicator LED, demonstrating correct system operation.

Overall, the experiment was successful. The system accurately extracted and amplified the small PPG signal, filtered it into a clean waveform, and converted it into a robust digital pulse train. Remaining discrepancies between measured and expected results can be explained by op-amp output swing limitations, tolerances in passive components, and finite sensor amplitude. Despite these factors, the circuit met its design goals and provided a reliable, real-time optical heart-rate detection capability.